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Centre number

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Candidate number

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Forename(s)

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# INTERNATIONAL AS CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Wednesday 22 May 2019 07:00 GMT Time allowed: 1 hour 30 minutes

## Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

## Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- All working must be shown.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

## Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

| For Examiner's Use |      |
|--------------------|------|
| Question           | Mark |
| 1                  |      |
| 2                  |      |
| 3                  |      |
| 4                  |      |
| 5                  |      |
| 6                  |      |
| 7                  |      |
| 8                  |      |
| <b>TOTAL</b>       |      |



Answer **all** questions in the spaces provided.

0 1

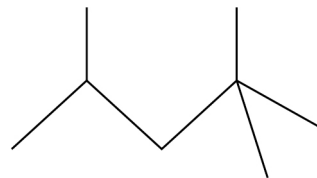
The skeletal formulas and boiling points of two structural isomers, **A** and **B**, are shown below.

**A**



boiling point = 125 °C

**B**



boiling point = 99 °C

0 1 . 1

Use molecular formulas to write an equation for the complete combustion of **A**.

[1 mark]

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0 1 . 2

Use IUPAC rules to name **B**.

[1 mark]

---

0 1 . 3

Explain why **A** has a higher boiling point than **B**.

[2 marks]

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0 1 . 4

A mixture of **A** and **B** can be separated in the laboratory.

Name the technique that can be used to separate **A** and **B**.

[1 mark]

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**0 1 . 5**

One molecule of an alkane with molecular formula  $C_{20}H_{42}$  is cracked. The products are tetramethylbutane, propene and ethene.

Use molecular formulas to deduce an equation for this reaction.

**[1 mark]**

---

**0 1 . 6**

The cracking reaction in Question **01.5** is carried out at a high temperature.

State **one** other condition needed.

**[1 mark]**

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**0 1 . 7**

Explain why the atom economy for the reaction in Question **01.5** is 100 %

**[1 mark]**

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**Question 1 continues on the next page**

**Turn over ►**

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | . | 8 |
|---|---|---|---|

Isomer **B** reacts with chlorine to produce a compound with  $M_r = 252.0$

This compound contains 38.10% carbon and 5.55% hydrogen by mass.

Calculate the empirical formula and the molecular formula of this compound.  
Show your working.

**[4 marks]**

Empirical formula \_\_\_\_\_

Molecular formula \_\_\_\_\_

|    |
|----|
| 12 |
|----|



**0 2 . 1** Van der Waals' forces exist between all molecules.

Explain how these forces arise.

**[3 marks]**

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**0 2 . 2** Draw a diagram to show the strongest intermolecular force between one ammonia molecule and one water molecule.

Include all partial charges and all lone pairs of electrons.

**[3 marks]**

6

**Turn over for the next question**

**Turn over ►**



|   |   |
|---|---|
| 0 | 3 |
|---|---|

An alcohol reacts with hot concentrated sulfuric acid to form a mixture of alkenes.

This alcohol also reacts with acidified potassium dichromate(VI) to form an organic compound that does **not** react with Tollens' reagent.

|   |   |   |   |
|---|---|---|---|
| 0 | 3 | . | 1 |
|---|---|---|---|

Explain how this information shows that this alcohol must be a secondary alcohol.

**[2 marks]**

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|   |   |   |   |
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| 0 | 3 | . | 2 |
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Pentan-2-ol can be dehydrated by reaction with hot concentrated sulfuric acid to form pent-2-ene and water.

Outline a mechanism for this reaction.

**[3 marks]**



0 3 . 3

Describe a test-tube reaction that can be used to show that pentan-2-ol has been dehydrated.  
Give the result.

**[2 marks]**

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0 3 . 4

Draw the skeletal formula of the product formed when pentan-2-ol reacts with acidified potassium dichromate(VI).

**[1 mark]**

0 3 . 5

Draw the displayed formula of the tertiary alcohol with molecular formula  $C_5H_{12}O$

**[1 mark]**

**0 4**

A factory uses propane as a fuel.

Mass spectrometry is used to monitor the oxides of nitrogen released in the waste gases from this factory.

**0 4 . 1**State the formula of an oxide of nitrogen with  $M_r = 44.0$ **[1 mark]**


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**0 4 . 2**Some precise atomic masses are given in **Table 1****Table 1**

| Element | Precise atomic mass |
|---------|---------------------|
| C       | 12.011              |
| H       | 1.008               |
| N       | 14.007              |
| O       | 15.999              |

Explain why precise atomic masses are needed to distinguish between molecular ion peaks of propane, carbon dioxide and the oxide of nitrogen in Question **04.1**

Use the data in **Table 1** to justify your answer.  
Show your working.

**[3 marks]**


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**0 4 . 3** Nitrogen monoxide (NO) is formed in internal combustion engines.

State **one** essential condition for the formation of NO in an engine.  
Write an equation to show the formation of NO in an engine.

**[2 marks]**

Condition \_\_\_\_\_

Equation \_\_\_\_\_

6

**Turn over for the next question**

**Turn over ►**



0 5

**F, G and H** are isomers of  $C_5H_{10}$ 

|                        |                     |                    |
|------------------------|---------------------|--------------------|
| $CH_2=C(CH_3)CH_2CH_3$ | $CH_3CH=CHCH_2CH_3$ | $CH_3CH=C(CH_3)_2$ |
| <b>F</b>               | <b>G</b>            | <b>H</b>           |

0 5 . 1

Use IUPAC rules to name **F**.

[1 mark]

---

0 5 . 2

**F** reacts with hydrogen bromide.Name and outline the mechanism for the reaction that forms the major product, **J**.Explain why **J** is the major product in this reaction.

[7 marks]

Name of mechanism \_\_\_\_\_

Mechanism

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Explanation \_\_\_\_\_

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0 5 . 3 **G** exists as *E* and *Z* isomers.

Draw the structure of the *E* isomer.

[1 mark]

0 5 . 4 Explain why **F** does **not** show *E–Z* isomerism.

[1 mark]

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0 5 . 5 Draw the structure of an isomer of  $\text{C}_5\text{H}_{10}$  that is **not** an alkene.

[1 mark]

Question 5 continues on the next page

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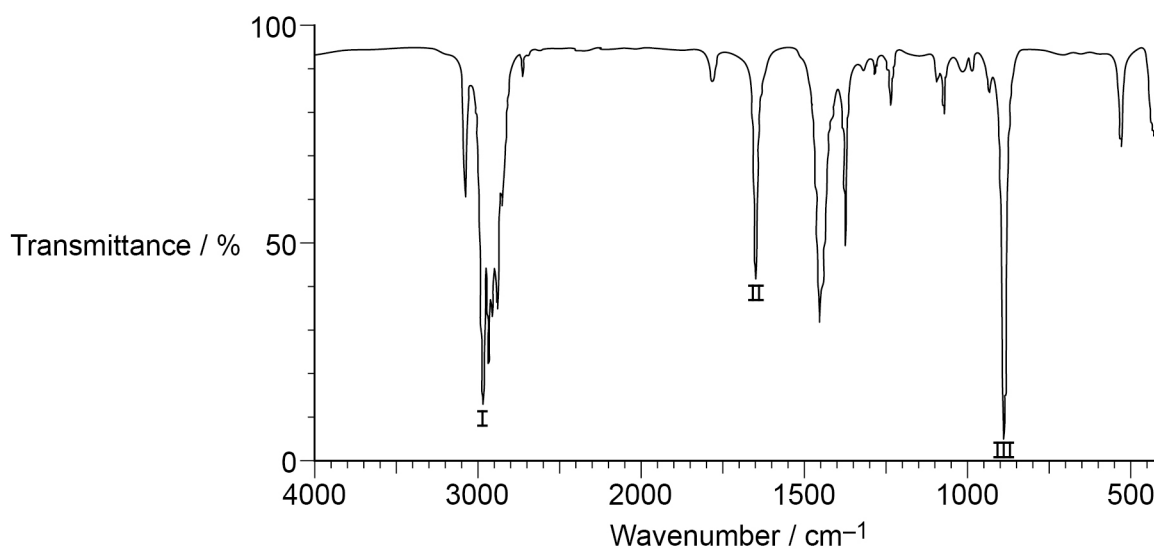
0 5 . 6

Draw one repeating unit of the polymer that can be formed from **H**,  $\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)_2$

[1 mark]

0 5 . 7

**Figure 1** shows the infrared spectrum of **H**.

**Figure 1**

Use **Table A** on the Data Sheet to identify the absorption that would **not** be present in the polymer.

Tick (✓) **one** box.

[1 mark]

**I**☐**II**☐**III**☐

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**0 6**

Hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) decomposes, in the presence of a catalyst, to form water and oxygen.

**0 6 . 1**

Write the equation for the decomposition of hydrogen peroxide.

**[1 mark]**

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**0 6 . 2**

Explain why the decomposition of hydrogen peroxide is a redox reaction.

Use oxidation states to justify your answer.

**[3 marks]**

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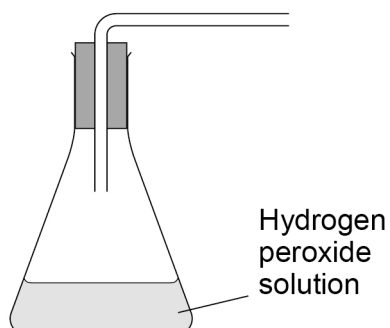
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**0 6 . 3**

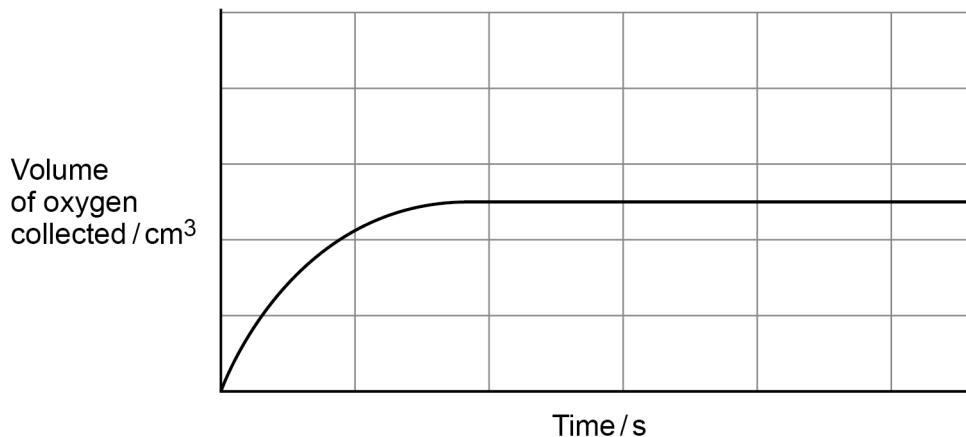
Complete the diagram to show the apparatus a student could use to collect the oxygen formed in this decomposition and measure its volume.

**[1 mark]**

5 cm<sup>3</sup> of a 1.0 mol dm<sup>-3</sup> solution of hydrogen peroxide decomposes, in the presence of a copper(II) oxide catalyst, at 298 K

**Figure 2** shows how the volume of oxygen collected changes with time.

**Figure 2**



**06.4** Explain why the curve is steepest at the start of the reaction.

**[2 marks]**

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**06.5** The experiment is repeated using 5 cm<sup>3</sup> of a 2.0 mol dm<sup>-3</sup> solution of hydrogen peroxide under the same conditions as in Question **06.4**

Draw a curve on **Figure 2** to show how the volume of oxygen collected changes with time in this experiment.

**[2 marks]**

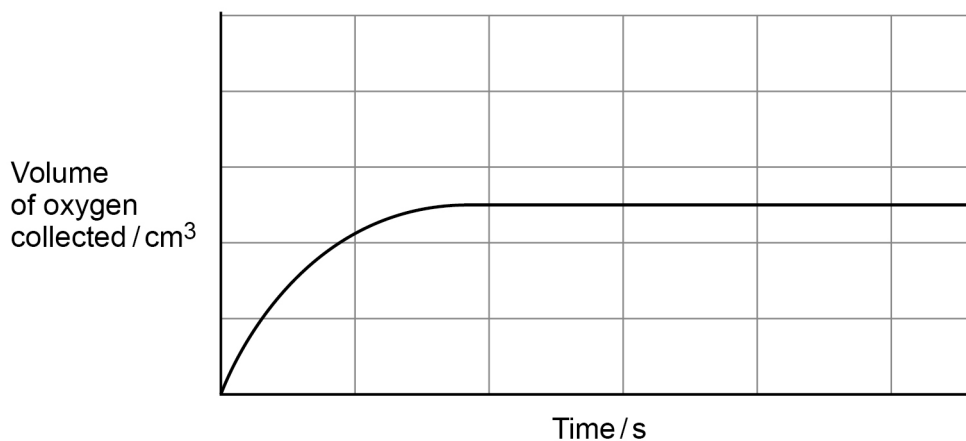
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0 6 . 6

**Figure 3** shows the volume of oxygen collected using  $5\text{ cm}^3$  of a  $1.0\text{ mol dm}^{-3}$  solution of hydrogen peroxide, in the presence of a copper(II) oxide catalyst, at  $298\text{ K}$

**Figure 3**



Potassium iodide is a more effective catalyst than copper(II) oxide.

The experiment is repeated using potassium iodide instead of copper(II) oxide.

Draw a curve on **Figure 3** to show how the volume of oxygen collected changes with time in this experiment.

**[2 marks]**





|   |   |   |   |
|---|---|---|---|
| 0 | 6 | . | 7 |
|---|---|---|---|

Calculate the maximum volume, in  $\text{cm}^3$ , of oxygen formed at 298 K and 100 kPa, in the reaction in Question **06.6**

The gas constant,  $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

**[3 marks]**

Volume \_\_\_\_\_  $\text{cm}^3$

|    |
|----|
| 14 |
|----|

**Turn over for the next question**

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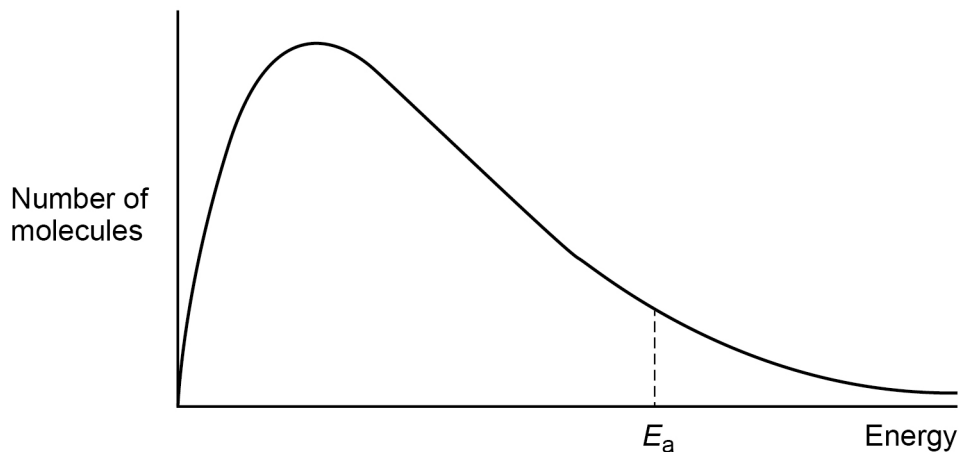


07

**Figure 4** shows the Maxwell–Boltzmann distribution for the molecules of a gas in a container of volume  $V$  and at temperature  $T$

The activation energy ( $E_a$ ) for its decomposition is shown.

**Figure 4**



07.1

Write the letter **M** on the appropriate axis of **Figure 4** to show the value of the most probable energy of the molecules of the gas at temperature  $T$

[1 mark]

07.2

The sample of gas at temperature  $T$  is placed in a container that has a volume of  $2V$

Explain why the Maxwell–Boltzmann distribution does **not** change for the gas in the new container.

[1 mark]

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07.3

State the effect, if any, of an increase in temperature on the value of  $E_a$  for this decomposition.

[1 mark]

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**0 8**

This question is about the reactions of halogenoalkanes.

**0 8 . 1**Halogenoalkanes react with reagent **Z** to form alcohols or alkenes depending on the conditions.Identify reagent **Z**.

State the condition(s) that favour the formation of each product.

Name the type of reaction in each case.

**[5 marks]**Reagent **Z** \_\_\_\_\_

Condition(s) to form alcohols \_\_\_\_\_

Type of reaction to form alcohols \_\_\_\_\_

Condition(s) to form alkenes \_\_\_\_\_

Type of reaction to form alkenes \_\_\_\_\_

**0 8 . 2**

Write an equation to show the reaction between chloroethane and an excess of ammonia.

Explain why this reaction is slower than the reaction between iodoethane and an excess of ammonia.

**[2 marks]**

Equation

\_\_\_\_\_

Explanation \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**7****END OF QUESTIONS**

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2 4



1 9 6 X C H O 2

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